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Navigating the Gen-AI Bifurcation: A Synergistic Framework for Occupation and Education

Summary

Addressing Gen-AI's disruption of labor and pedagogy, this study introduces the Gen-AI-Occupation-Education Synergistic Framework to quantify socioeconomic impacts and optimize institutional strategies.

First, we formulate a **Jump-Diffusion AI Trajectory Model (JDAT)**, synthesizing a **non-homogeneous Poisson jump process** to capture stochastic breakthroughs and **logistic implementation lags** to model industrial absorption. Calibrated via **MLE** on global LLM datasets, the JDAT establishes capability benchmarks across language, reasoning, and multimodal dimensions within a 95% confidence interval.

Subsequently, the **Career Future Exploration (CFE)** model projects occupational trajectories by mapping Gen-AI latent capabilities through **AHP** matrices into task-level **logistic substitution and augmentation rates**. These micro-impacts are integrated into a **Cobb-Douglas production function** to derive **Marginal Productivity of Labor (MPL)** equilibria. Simulation results for 2024–2033 indicate a profound **structural bifurcation**:

- **Actuaries (STEM):** Experience a 86.31% augmentation surge and 42.59% employment expansion, driven by high Gen-AI-capitalization potential.
- **Chefs (Trades):** Exhibit high **resilience** with stable employment; sensory-physical tasks remain a "moat" against automation (substitution $k = 16.98\%$).
- **Graphic Designers (Arts):** Suffer extreme disruption (substitution $k = 121.22\%$), yet high-tier roles command a 11.83% wage premium due to **scarcity-driven demand for "Originality."**

Strategic institutional responses are optimized via the **Education Policy Formulation (EPF)** model, utilizing **SQP** to maximize graduate utility under constraints. Optimal Gen-AI openness (ϵ) is determined at **0.7651** for Actuaries (pivoting to "Tech-Dev"), **0.1014** for Chefs (reinforcing "Sensory Mastery"), and **0.6592** for Designers (focusing on "Aesthetic Curation").

Finally, **Sobol global sensitivity analysis** validates model robustness (S_{Ti} analysis). We conclude that the optimal pedagogical trajectory necessitates a strategic shift from generic vocational training to the cultivation of **Human Resilience Assets** and **AI-Native Proficiencies** to navigate the profound shocks of the Gen-AI era.

Keywords: Generative AI;Task-based Model; Jump-Diffusion Process;Job Polarization;Multi-objective Optimization;Education Reform.

Contents

1 Introduction

1.1 Background

Within a few years, Gen-AI has evolved from a nascent tool to a cognitive engine capable of generating high-quality code, images and insights with superhuman speed. Distinct from previous mechanical automation supplanting repetitive physical labor or other kinds of AI (conventional analytical AI etc.) , Gen-AI primarily challenges the "human monopoly" on creativity and complex decision-making.

Research suggests the profound but asymmetrical impact of Gen-AI on the future of work: while some sectors face a "substitution crisis", others experience "augmented growth". For post-secondary educational institutions ,this shift necessitates an urgent pedagogical transformation. Despite some policy adjustments made , balancing technological proficiency with human-centric value remains a critical challenge. Therefore, it is an imperative for institutional leaders to formulate comprehensive policies to ensure the long-term employability and societal value of their graduates in a Gen-AI-ubiquitous world.

1.2 Restatement of the Problem

To provide a roadmap for Post-Secondary Institutions in the era of Gen-AI, we aim to address the following tasks:

- **Task 1: Explore the future of Careers.** Select one representative profession from each of the three categories: STEM, Trades, and Arts. Design a data-informed model to explore the future of each profession, identifying key drivers and the expected impacts of Gen-AI on human workload.
- **Task 2: Formulate Institutional Recommendations.** Identify three specific post-secondary institutions for each profession. Provide strategic guidance for institutional leaders to recalibrate study programs and other policy reforms.
- **Task 3: Perform holistic policy appraisal and assess strategic extensibility.** Incorporate externalities beyond employability to assess and refine the proposed policies. Evaluate the generalization and robustness of the model.

1.3 Literature Review

The research landscape and its implications for our modeling framework are visually summarized in Figure 1.

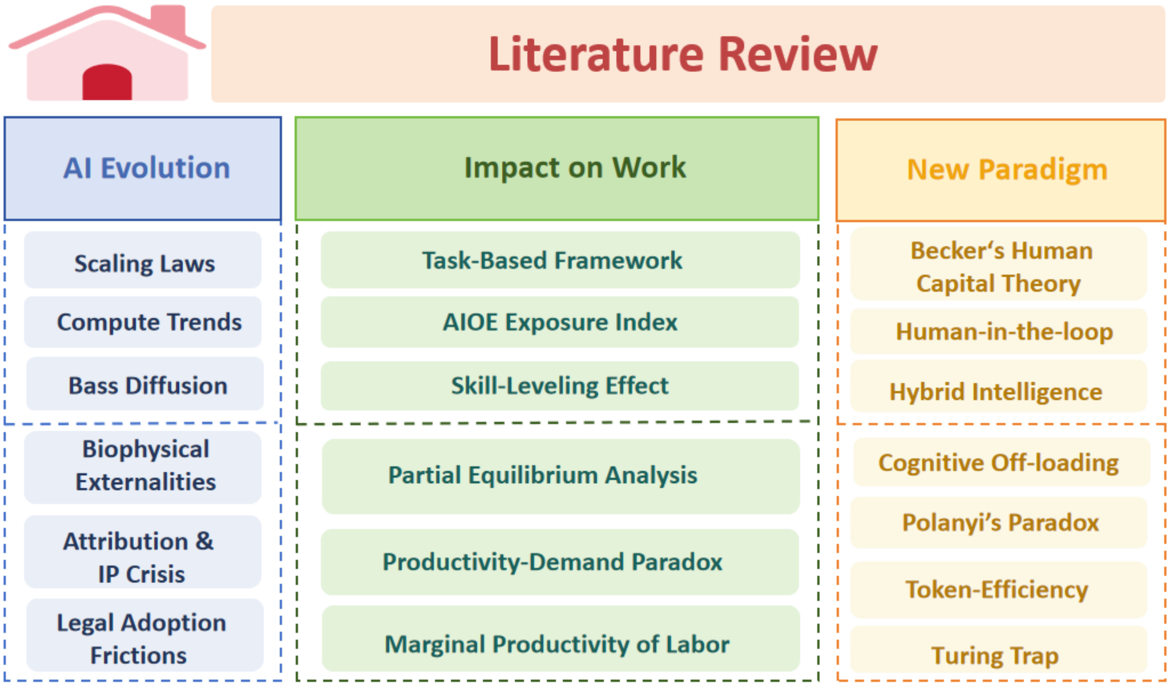


Figure 1: Literature Review

1.4 Our Work

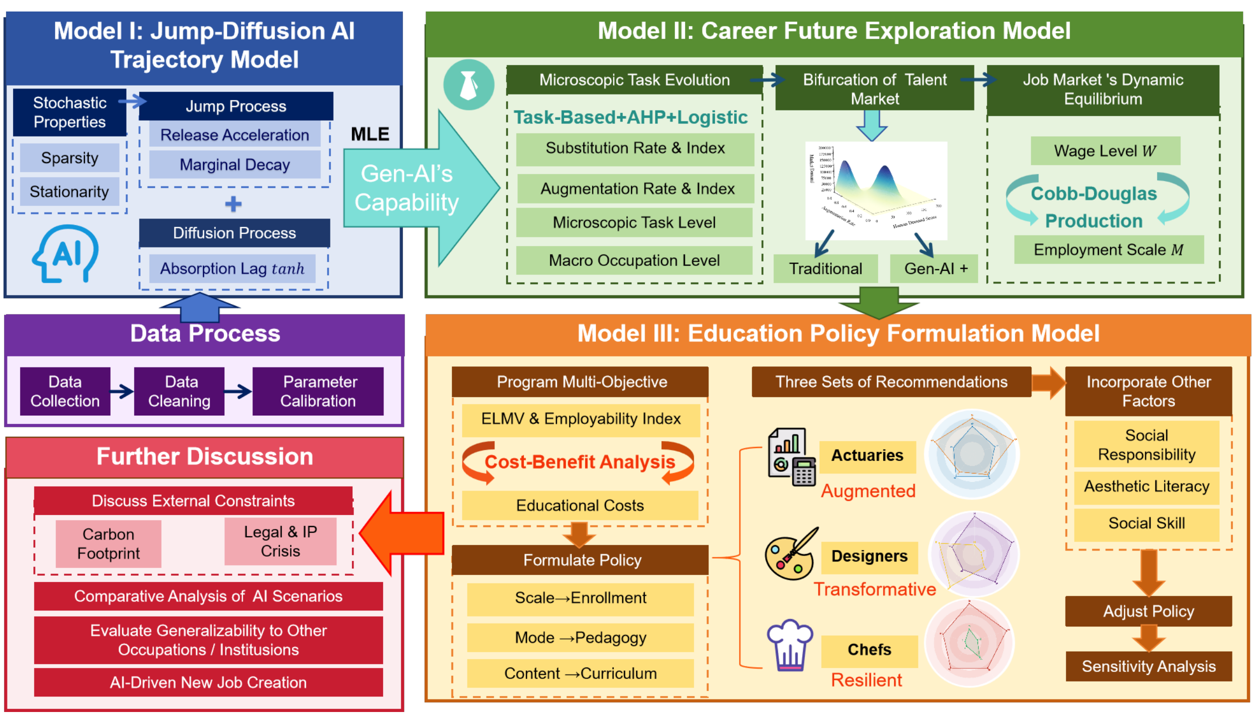


Figure 2: Our Work

2 Model Preparation

2.1 Assumptions and Justifications

To transform the complex socio-technical dynamics of Gen-AI into a manageable mathematical framework, we establish the following assumptions:

Assumption 1: Systematic Trajectory and Environmental Stability. Gen-AI evolves following a systematic path characterized by *Scaling Laws*, while the broader socio-economic background remains relatively stable.
⇒ *Justification:* This isolates Gen-AI as the primary driver, providing a clear baseline to assess its net impact without interference from unrelated macroeconomic noise.

Assumption 2: Commensurability and Linear Mapping of Competencies. The competencies required for a specific profession and the functional capabilities of Gen-AI are both quantifiable and comparable through a specific linear transformation.
⇒ *Justification:* Established research on "task automation" frequently utilizes indexed scoring systems, validating that measuring occupational exposure through weight-based linear combinations is a robust and academically recognized methodology.

Assumption 3: Task-level Granularity and Additivity. The total Gen-AI exposure of a profession is the weighted sum of its constituent tasks.
⇒ *Justification:*This aligns with the **Task-Based Framework**, enabling the transition from qualitative descriptions to quantitative modeling across diverse sectors.

Assumption 4: Quantifiable and Quadratic Reform Costs. The expenditures incurred by institutions for curriculum restructuring and Gen-AI management are quantifiable, with the total cost being proportional to the square of the reform intensity.
⇒ *Justification:* This accounts for the non-linear escalation of costs as reforms intensify, reflecting the increasing systemic friction and resource demands of deep institutional change.

2.2 Notations

The notations for the model under study are shown below in Table 1.

Table 1: Notation and Definitions

Symbol	Description
t	Time variable (base year 2024, $t = 0$)
$\vec{a}(t)$	Vector of aggregate Gen-AI capabilities
$\vec{b}(t)$	Vector of intrinsic human labor capabilities
$\vec{\kappa}(t)$	Vector of Gen-AI substitution rates per task
$\vec{\zeta}(t)$	Vector of Gen-AI augmentation rates per task

Continued on next page

Symbol	**Description** (continued)
$k(t)$	Gen-AI Substitution Index for an occupation
$z(t)$	Gen-AI Augmentation Index for an occupation
$M(t)$	Employment scale (market size)
$W(t)$	Average wage level
$\vec{X}(t)$	Vector of graduates' cultivated proficiency
P	Employability Index (hiring probability)
C	Total cost function of PSIs
U	Total objective function of PSIs

Note: Throughout this paper, "AI" refers exclusively to "Gen-AI" and "PSIs" refers to "Post-Secondary Institutions".

A Note on Terminology

To ensure analytical clarity, we establish a hierarchy for categorization:

Table 2: Terminology and Definitions

Terminology	**Description**
Career	The macro-level, long-term trajectory of professional evolution
Occupation	The meso-level classification of professional roles (e.g., Actuary, Chef)
Job	The micro-level unit for analyzing wage dynamics and labor demand
Profession	The institutional domain for higher educational policy

We synthesize data from three core subjects: Gen-AI, occupations (& labor market), and PSIs. To ensure empirical validity, detailed dataset descriptions and authoritative provenances are provided directly within the modeling procedures.

2.3 Career Selection and Capability Vectorization

To maximize our model's explanatory power, we selected three occupations that represent the full spectrum of Gen-AI's impact—a strategy aimed at capturing **Maximal Explanatory Variance** rather than predicting uniform growth. Our archetypes are:

1. **Actuaries (STEM):**High Cognitive Exposure, High Resilience.
2. **Chefs (Trade):** Low Cognitive Exposure, High Extreme Resilience.
3. **Graphic Designers (Arts):** High Substitution/Augmentation Dualism.

For a rigorous and consistent approach :

- **Human Capability Vector ($\vec{b}(t)$):** We utilized the ***O*NET Database***[1] to identify and quantify the ***top five most critical skill dimensions*** for each target occupation, ensuring our human input accurately reflects the core value proposition of the labor.
- **AI Capability Vector ($\vec{a}(t)$):** $\vec{a}(t)$ is a four-dimensional vector synthesizing classifications from the ***HAI AI Index Report***[2], the ***AIOE Index***[3], and the core needs of our three professions.

3 Model I:The Jump-Diffusion AI Trajectory Model(JDAT)

3.1 Model Foundations: The Stochastic Properties

To gain insight into the complex effects of Gen-AI on the future of careers, we must first model the evolution of Gen-AI. We characterize the cumulative potential capability of Gen-AI as a jump process, according to three properties:

- **Sparsity & Stationarity:** Gen-AI progress is driven by discrete, high-impact events (e.g., GPT-4).
- **Empirical Alignment:** Analysis of frontier model releases (2019-2024) shows that inter-arrival times ΔT follow an **Exponential Distribution**. This mathematically confirms that the release counting process is a Poisson process with intensity λ.

Consequently, the total potential capability vector $\vec{A}(t)$ is defined as the summation of all preceding stochastic shocks:

$$\vec{A}(t) = \vec{A}(0) + \sum_{i=1}^{N(t)} \vec{X}_i \tag{1}$$

Where random variables $\vec{X}_i$ represents the **magnitude vector** of the technological jump induced by the i-th release.

3.2 Model Formulation: The Jump-Diffusion Logic

3.2.1 Jump Process

Guided by empirical observations from **Villalobos et al.**[4] and **Sevilla et al.**[5], contemporary Gen-AI evolution is characterized by two opposing forces: the **"Data Wall"** (marginal decay of individual leaps due to high-quality data exhaustion) and the **"Compute Surge"** (exponential increase in release frequency driven by massive hardware investment).

Thus, the time-varying intensity $\lambda(s)$ and the mathematical expectation of decaying magnitude $E[\vec{X}(s)]$ are defined as follows:

- **Release Acceleration (α):** Reflecting the surge in compute-driven iterations, the arrival intensity follows:

$$\lambda(s) = \lambda_0 e^{\alpha s} \tag{2}$$

- **Marginal Decay (δ):** Reflecting data scarcity and diminishing returns of scaling, the expected magnitude of a single leap follows:

$$E[\vec{X}(s)] = \vec{X}_0 e^{-\delta s} \tag{3}$$

3.2.2 Diffusion Process

Considering the **Absorption Lag** (the delay between lab breakthrough and industrial penetration), we multiplied each term within the summation of formula (1) by an implementation conversion function $p_a(\tau)$ based on **Bass Diffusion Theory** [6]. We employ a hyperbolic tangent to characterize the transition from experimental prototype to full-scale adoption:

$$p_a(\tau) = \varepsilon(\tau) \tanh\left(\frac{k\tau}{2}\right) \tag{4}$$

where τ denotes the time elapsed since model release, and $\varepsilon(t)$ is the unit step function.

Based on authoritative datasets, we obtain the values of the core parameters through **Non-linear Regression**; the parameter values and fitting results are as follows:

3.3 Model Results & Implications: Expected Capability Range

To forecast the long-term Gen-AI trajectory, we transition from discrete stochastic shocks to continuous approximations. Applying **Campbell's Theorem** yields the expected realized capability $E[\vec{A}(t)]$ and its variance $Var[\vec{A}(t)]$, convolving exponential advancement with absorption lags:

$$\mathbb{E}[\vec{A}(t)] = \vec{A}(0) + \lambda_0 \vec{X}_0 \int_0^t e^{(\alpha-\delta)s} \cdot \tanh\left(\frac{k(t-s)}{2}\right) ds \tag{5}$$

$$\mathrm{Var}[\vec{A}(t)] = \lambda_0 E[\vec{X}^2(0)] \int_0^t e^{(\alpha-2\delta)s} \cdot \tanh^2\left(\frac{k(t-s)}{2}\right) ds \tag{6}$$

We calibrated the model using Non-Linear Least Squares against benchmarks from the *HAI AI Index Report 2025* (Language, Audio, Visual, Reasoning)[2]. As shown in Figure 3, the model demonstrates superior goodness-of-fit across all dimensions.

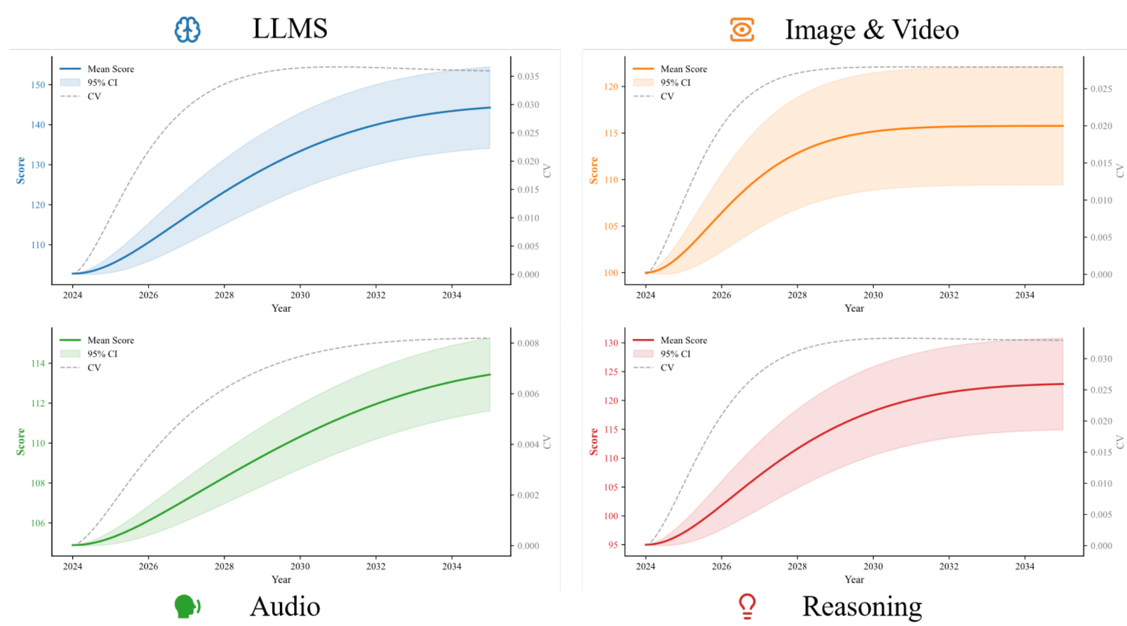


Figure 3: Gen-AI Trajectory

All Gen-AI dimensions follow S-curve trajectories, with LLMs and Reasoning leading in magnitude while Image & Video reaches saturation earliest near 2030. To further quantify risk and predictability, we construct prediction intervals using the **Asymptotic Normal Approximation**. At a confidence level of 95% , the capability envelope is defined as:

$$\vec{A}_{\pm}(t) = \mathbb{E}[\vec{A}(t)] \pm z_{0.975} \cdot \sqrt{\mathrm{Var}[\vec{A}(t)]} \tag{7}$$

This framework yields three distinct scenarios for Gen-AI evolution:

Optimistic Estimate ($\vec{a}_{+}(t)$): The upper critical limit (97.5^{th} percentile).

Pessimistic Estimate ($\vec{a}_{-}(t)$): The lower critical limit (2.5^{th} percentile).

Neutral Estimate ($\vec{a}(t)$): The expected value $\mathbb{E}[\vec{A}(t)]$, serving as the central baseline for all subsequent mathematical derivations and predictive analysis.

4 Model II:The Career Future Exploration Model (CFE)

4.1 Model Preparation

To systematically evaluate the heterogeneous effects of Gen-AI on labor, we categorize the tech-human interaction into three distinct dimensions:

- **Substitution:** Displacement of routine tasks due to cost-efficiency;

- **Augmentation:** Productivity enhancement via complementary intelligence;
- **Resilience:** Immunity to automation due to physical embodiment or ethical requirements.

To bridge the gap between Gen-AI's latent capabilities and job requirements, we introduce the **Mapping Matrix** E. This matrix projects the Gen-AI Capability Vector $\vec{a}(t)$ (from the JDAT model) into the space of occupational task requirements. The resulting Gen-AI Occupational Competence vector $\vec{A}_{\text{impact}}(t)$ is given by:

$$\vec{A}_{\text{impact}}(t) = E\vec{a}(t) \tag{8}$$

where E is calibrated using task-level data from **Felten et al. (2021)**[3] and refined via the **Analytic Hierarchy Process (AHP)**. The full computational procedure of the AHP is omitted due to space constraints.

In our model, certain terms in E are normalized to zero to account for this Resilience Moat. Consequently, the subsequent analytical framework concentrates exclusively on Substitution and Augmentation.

4.2 Microscopic Task Evolution Model

To better quantify the substitution and augmentation impacts, we introduce two key rates: the **Substitution Rate ($\vec{\kappa}(t)$)** and the **Augmentation Rate ($\vec{\zeta}(t)$)**.

4.2.1 Substitution Rate

The total capability requirement for a task, $\vec{f}(t)$, is fulfilled by the joint contribution of human proficiency $\vec{b}(t)$ and Gen-AI-driven capability $E\vec{a}(t)$:

$$\vec{b}(t) + E\vec{a}(t) = \vec{f}(t) \tag{9}$$

To account for the continuous upskilling required for competitiveness, we model $\vec{f}(t)$ as a linear regression fit of the O*NET dataset[1] and thus derive $\vec{b}(t)$ from the equation above.

Obviously, the substitution effect is intrinsically non-linear, characterized by a **"winner-take-all"** effect. Gen-AI integration remains negligible until its performance surpasses the human proficiency baseline, at which point capital interests drive a rapid, large-scale displacement even if the Gen-AI's marginal superiority is slight.

To capture these dynamic features, we model the **Substitution Rate** $\vec{\kappa}(t)$ using a **Logistic Function**. Specifically, the i^{th} component of $\vec{\kappa}(t)$ is formulated as:

$$\kappa_i(t) = \frac{1}{1 + e^{-\left(\frac{(E\vec{a}(t))_i}{b_i(t)} - 1\right)}} \tag{10}$$

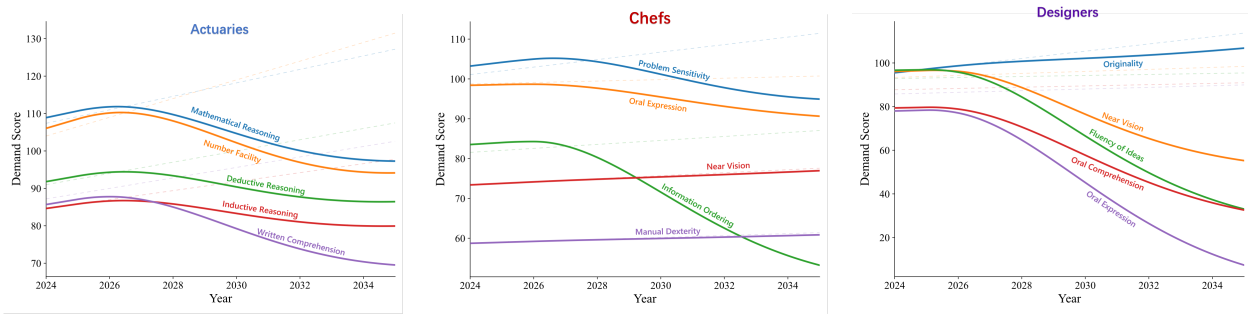
4.2.2 Augmentation Rate

Beyond direct substitution, Gen-AI serves as an **efficiency multiplier**. We define the **Augmentation Rate** $\vec{\zeta}(t)$, representing the marginal time-saving ratio or productivity gain of "AI-equipped" labor compared to a "human-only" state, where each component $\zeta_i(t)$ quantifies the enhancement specific to the i^{th} task.

Similarly, as workers only employ Gen-AI tools when the utility reaches a certain tipping point, we formulate the i^{th} component of the Augmentation Rate $\vec{\zeta}(t)$ as:

$$\zeta_i(t) = \frac{1}{1 + e^{-C_i \cdot (E\vec{a}(t))_i}} \tag{11}$$

We utilized the **OECD AI Employment Outlook 2025**[7] dataset for the calibration of C_i and obtained the following results for the three selected professions:



(a) Actuaries (STEM) (b) Chefs (Trades) (c) Designers (Arts)

Figure 4: Comparison of Augmentation Rates across Three Occupations

Across all sectors, AI-substitutable cognitive tasks decline significantly. Chefs maintain stability in physical assets like Manual Dexterity, whereas Designers exhibit a unique upward pivot in Originality, transforming it from a routine skill into a high-premium scarcity.

4.3 Macro Occupation Evolution Model

4.3.1 Occupational Substitution and Augmentation Index

To forecast developmental trends from a macro perspective, we aggregate the task-level rates into two weighted indexes, defined as the product of the **Task Significance Vector** $\vec{w}$ and their respective task rates:

- **Occupational Substitution Index ($k(t)$):**

$$k(t) = \vec{w}^T \vec{\kappa}(t) \tag{12}$$

- **Occupational Augmentation Index ($z(t)$):**

$$z(t) = \vec{w}^T \vec{\zeta}(t) \tag{13}$$

Where $\vec{w}$ is the Task Significance Vector, derived by normalizing the product of Importance and Level scores from the O*NET dataset[1], satisfying $||\vec{w}||_1 = 1$.

Using the task weights for the three selected professions, Figures 8 and 9 showcase the evolution of calculated macro indices over the next decade:

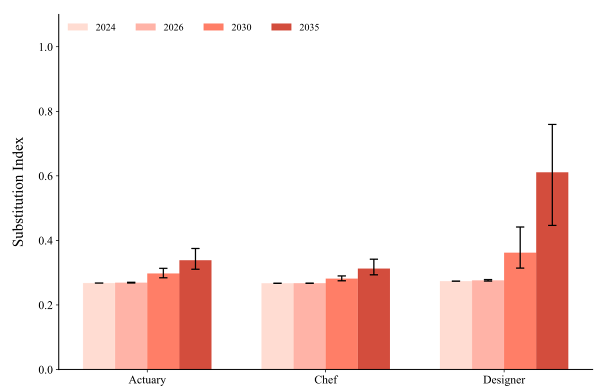


Figure 5: Substitution Index

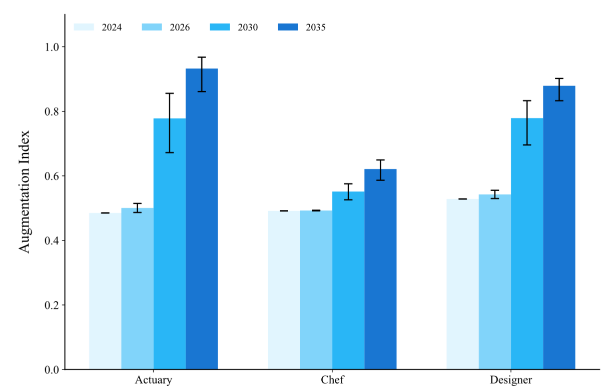


Figure 6: Augmentation Index

4.3.2 The Bifurcation of the Labor Market

Analysis of the resulting indices $k(t)$ and $z(t)$ for the three archetypes reveals a structural shift consistent with **Job Polarization** [8]. As shown in Figure 7, the labor market is bifurcating into two distinct high-value clusters, driven by the divergence of substitution and augmentation impacts:

- **The Gen-AI+ Job (High $z(t)$):** These roles leverage Gen-AI as a force multiplier to achieve hyper-productivity. Value creation is anchored in the deep fusion of human curation and machine efficiency.

- **The Traditional Job (Low $\vec{k}(t)$):** This category encompasses roles where substitution is hindered by **Non-Technical Friction**, such as artistic purism (e.g., non-AI illustration), industry inertia, or delayed capital transition. These roles persist in a "traditionalist" market, decoupled from the Gen-AI frontier due to ethical preferences or systemic lag.

This structural shift creates a **"Mediocrity Trap,"** as workers with balanced but low-magnitude scores in both $k(t)$ and $z(t)$ find themselves in the **"Valley of Obsolescence."** These mid-tier practitioners face the highest displacement risk, being neither cost-competitive with Gen-AI nor qualitatively superior to the elite clusters.

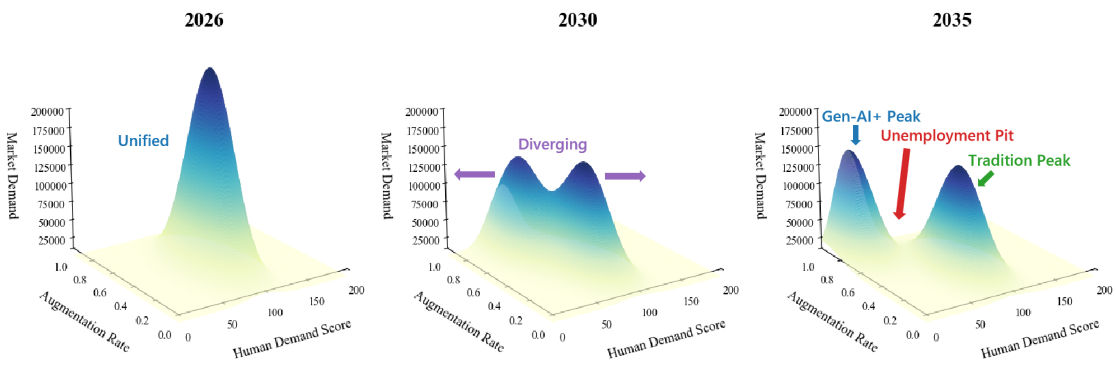


Figure 7: The Bimodal Landscape of Future Employment

- **Type:** 3D Surface Plot (Probability Density / Market Value Landscape).
- **Axes:** X-axis represents **AI Integration Level** ($z(t)$), Y-axis represents **Traditional Human Skill Depth** (Resilience), and Z-axis represents **Market Demand/Viability**.
- **Visual:** The plot clearly shows two distinct peaks: the Gen-AI+ Peak (High z, High Demand) and the Resilience Peak (High Human Skill, High Demand). Between them lies a deep **"Valley of Obsolescence"** representing the marginalized middle.

4.4 Job Market's Dynamic Equilibrium Model

4.4.1 Employment Scale

Integrating the Gen-AI's dual impact and **Harris-Todaro Model**[9] of labor migration, the employment scale ($M_i(t+1)$) of each sector is governed by two countervailing economic forces: the **Direct Gen-AI Impact** and the **Wage-Driven Labor Reallocation**.

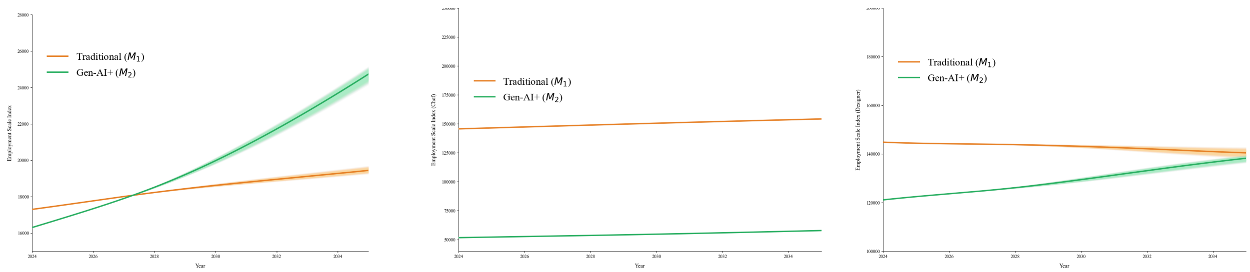
$$M_1(t+1) = M_1(t)(1 + \rho - \delta_1 k(t)) - \eta(W_2(t) - W_1(t)) \quad (14)$$
$$M_2(t+1) = M_2(t)(1 + \rho + \delta_2 z(t)) + \eta(W_1(t) - W_2(t)) \quad (15)$$

Note: Subscripts 1 and 2 denote Traditional and Gen-AI+ jobs, respectively. Subsequent notations follow this convention.

- **Exogenous Growth Rate (ρ):** The natural expansion of the industry, calibrated using Bureau of Labor Statistics (BLS) datasets[10].
- **AI Elasticity of Employment (δ_i):** δ_1 quantifies the displacement impact for traditional jobs while δ_2 quantifies the capitalization for Gen-AI jobs, estimated using [10] and **The Future of Jobs Report 2025**[11].
- **Labor Mobility Coefficient (η):** The sensitivity of the workforce to wage differentials. A higher η indicates lower labor market friction, estimated based on [11] [12].

W_i denotes the wage level, to be detailed hereafter. Initial values for $M_1 + M_2$ are sourced directly from the BLS dataset, while their ratio is estimated based on [11].

The derived employment scale changes for the three occupations are shown in **Figure 8**.



(a) Actuaries (STEM) (b) Chefs (Trades) (c) Designers (Arts)

Figure 8: The Bifurcation of the Talent Market: Employment Scale Evolution

Industry-Internal Comparison (Intra-Sectoral): Shows the migration trend: the decline of $M_1(t)$ (Traditional Job) and the growth of $M_2(t)$ (Gen-AI+ Job) within the same industry (e.g., Actuaries).

Inter-Industry Comparison (Inter-Sectoral): Compares the total employment trend $M_{total}(t) = M_1(t) + M_2(t)$ across the three archetypes, illustrating how Job Polarization leads to aggregate job growth in some sectors (e.g., STEM) but contraction in others (e.g., Trade).

4.4.2 Wage Level

To translate Gen-AI breakthroughs into economic outcomes, we employ the **Cobb-Douglas Production Function**[13], ideal for long-term sectoral modeling. We divide the total sectoral output $Y(t)$ into two components: the **Traditional ($Y_1(t)$)** and the **Gen-AI+ ($Y_2(t)$)** sectors, where $Y(t) = Y_1(t) + Y_2(t)$.

Each sector follows the production frontier:

$$Y_i(t) = G_i(t) M_i(t)^{\omega} K_i^{1-\omega} \tag{16}$$

- **Capital Input (K_i):** Assumed to be quasi-fixed, according to Assumption 1.
- **Output Elasticity (ω):** Set based on **OECD estimations**[7], representing the empirical labor share of GDP in developed economies.

Total Factor Productivity (TFP) ($G_i(t)$) is Modeled linearly against Gen-AI indices to internalize the **Solow Residual**, quantifying Gen-AI's contribution.

- **Traditional Sector:** Gen-AI induces "Schumpeterian Creative Destruction," where labor contribution to TFP is gradually eroded: $G_1(t) = G_0(1 - \psi_1 k(t))$

- **Gen-AI+ Sector:** TFP experiences endogenous growth through human-AI collaboration: $G_2(t) = G_0(1 + \psi_2 z(t))$

4.4.3 Wage Dynamics and Equilibrium

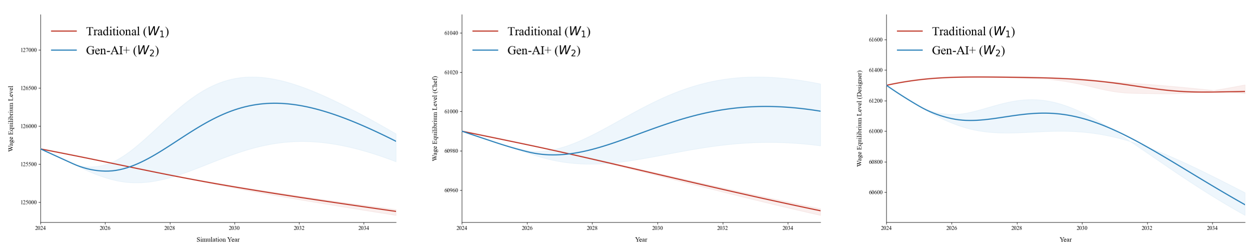
In a competitive labor market, wages are determined by the **Marginal Revenue Product of Labor (MRPL)**. Taking the partial derivative $\frac{\partial Y}{\partial M_i}$ yields the equilibrium wage for each sector. Assuming both sectors share a baseline wage W_0 at $t = 0$, the wage evolution trajectories are:

$$W_1(t) = W_0 \frac{(1 - \psi_1 k(t)) M_1(t)^{\omega - 1}}{(1 - \psi_1 k(0)) M_1(0)^{\omega - 1}} \tag{17}$$

$$W_2(t) = W_0 \frac{(1 + \psi_2 z(t)) M_2(t)^{\omega - 1}}{(1 + \psi_2 z(0)) M_2(0)^{\omega - 1}} \tag{18}$$

The coefficients ψ_1 and ψ_2 are estimated based on [14].

The resulting wage evolution for the three occupations is shown in **Figure 7**.



(a) Actuaries (STEM) (b) Chefs (Trades) (c) Designers (Arts)

Figure 9: Wage Evolution Trajectories across Three Occupations

Industry-Internal Comparison (Intra-Sectoral): Shows the growing wage differential between the **Traditional** ($W_1(t)$) and **Gen-AI+** ($W_2(t)$) versions of the same job within the Actuaries and Graphic Designers archetypes.

Inter-Industry Comparison (Inter-Sectoral): Compares the average wage trend $W_{avg}(t)$ across the three different archetypes (STEM, Trade, Arts), highlighting how Job Polarization affects high- and low-skill jobs differently.

5 Model III : The Education Policy Formulation Model (EPF)

5.1 Multi-Objective Policy Optimization

According to **Becker's Human Capital Theory**[15], PSIs' primary objective is to maximize the **Expected Labor Market Value (ELMV)** of their graduates, while minimizing policy implementation costs (e.g., curriculum redesign, faculty retraining, etc.).

5.1.1 Quantification of ELMV and Employability Index

A. Labor Demand Aspect

From the labor market perspective, the occupational requirements are represented by the vector:

$$\vec{x}(t) = (\vec{b}(t)^T, \vec{\zeta}(t)^T)^T \tag{19}$$

In light of the market bifurcation established in Section 4.3.2, labor demand has diverged into two distinct requirements for graduates. We assume traditional industry requirements maintain relative stability $\vec{x}(0)$, whereas Gen-AI+ sectors demand an evolving $\vec{x}(t)$. All components are normalized at $t = 0$ for longitudinal comparability.

B. Labor Supply Aspect

In view of the current market bifurcation, Prospective practitioners face a strategic trade-off between two developmental paths. Similarly, constrained by limited instructional time, educators must navigate this trade-off when designing academic programs:

- **The Traditional Path:** Maintaining the baseline foundational proficiency $\vec{b}(0)$.
- **The GenAI+ Path:** Cultivating the Gen-AI-integrated capability vector $\vec{b}(t)$.

Correspondingly, we quantify the expected proficiency metrics of students developed by PSIs as follows.

- **Cultivated Traditional Proficiency $\vec{B}(t)$:** Instituting pedagogical reforms shifts the skill level from the baseline $\vec{b}(0)$ to $\vec{B}(t)$. Entry Barriers exist for highly specialized domains, requiring $\vec{B}_i \geq \vec{B}_{i,\min}$ or $\vec{B}_i = 0$.
- **Cultivated Gen-AI-Enabled Proficiency $\vec{Z}(t)$:** $\vec{Z}(t)$ is subject to an Gen-AI literacy threshold $\vec{Z}_i \geq \vec{Z}_{i,\min}$ (or $\vec{Z}_i = 0$) for meaningful use.

Note: Unless otherwise specified, the time parameter t consistently aligns with the duration of the respective academic program.

Thus, the **Comprehensive Capability Vector** of a graduate is represented as a concatenated column vector:

$$\vec{X}(t) = (\vec{B}(t)^T, \vec{Z}(t)^T)^T \tag{20}$$

5.1.2 Employability Index

To quantify the probability of securing employment, we define the **Employability Index** $P_i(t)$ as a function of the **Weighted Euclidean Distance** between graduate skills $\vec{X}(t)$ and market demands $\vec{x}(t)$:

- **Traditional Job ($P_1(t)$):**

$$P_1(t) = k_0 M_1(t) \frac{1}{1 + \sqrt{\frac{1}{2}(\vec{x}(0) - \vec{X}(t))^T W (\vec{x}(0) - \vec{X}(t))}} \tag{21}$$

- **Gen-AI+ Job ($P_2(t)$):**

$$P_2(t) = k_0 M_2(t) \frac{1}{1 + \sqrt{\frac{1}{2}(\vec{x}(t) - \vec{X}(t))^T W (\vec{x}(t) - \vec{X}(t))}} \tag{22}$$

where:$W = \mathrm{diag}\{w_1, w_2, \ldots, w_n, w_1, w_2, \ldots, w_n\}$. The Denominator represents the **"Skill Gap"** distance. As the gap increases, the hiring propensity decays. k_0 is a normalization constant calibrated such that $P(0) = P_1(0) + P_2(0) = 1$ at $t = 0$.

5.1.3 AI Adoption Intensity

Institutions must navigate the strategic dilemma of the whether to encourage the integration of Gen-AI. We quantify this as the Gen-AI Adoption Intensity $\epsilon \in [0, 1]$. On one hand, the adoption of Gen-AI liberates students from routine, low-value tasks, thereby enhancing learning efficiency. This, in turn, expands the theoretical maximum proficiency a student can attain:

$$(1 + k_\epsilon \epsilon)||\vec{X}(0)||_1 \geq ||\vec{X}(t)||_1 \tag{23}$$

where k_ϵ represents the **Learning Efficiency Gain** facilitated by Gen-AI tools (e.g., accelerated coding or literature synthesis), obtained from [16].

Conversely, Gen-AI reliance may trigger skill atrophy and deep-learning degradation. To account for these setbacks and the capital expenditure of Gen-AI integration, we define an Adoption Overhead associated with ϵ:

$$C_\epsilon(t) = q\epsilon^2 \tag{24}$$

5.1.4 Quantification of Educational Costs

The transition of the educational paradigm incurs two types of costs: **Direct Reform Costs ($C_{\vec{X}}$)** and **Implicit Gen-AI-Policy Costs (C_ϵ and C_Θ).**

The cost of upgrading student competencies is modeled using a **Quadratic Form**[17]:

$$C_{\vec{X}}(t) = (\vec{X}(t) - \vec{X}(0))^T \boldsymbol{L} (\vec{X}(t) - \vec{X}(0)) \tag{25}$$

where $\boldsymbol{L}$ is a diagonal **Cost-Sensitivity Matrix**, calibrated based on a comprehensive assessment of the scale and financial capacity of the involved institutions. Each element L_{ii} represents the **Marginal Cost Escalation** of increasing one unit of proficiency.

Similarly,

$$C_{\Theta}(t) = l_{\Theta}\left(\Theta(t) - \Theta(0)\right)^2 \tag{26}$$

where $\Theta(t)$ denotes the aggregate of holistic competencies (e.g., social, aesthetic, and moral development) not directly linked to vocational utility, determined by the specific curriculum of the chosen program.

The **Total Educational Cost** is thus:

$$C(t) = C_{\vec{X}}(t) + C_{\epsilon}(t) + C_{\Theta}(t) \tag{27}$$

5.1.5 Multi-Objective Optimization

Institutions seek to maximize a **Social Utility Function** U, balancing the **Expected Labor Market Value (ELMV)** of all graduates against fiscal constraints:

$$\max U = k_1[P_1(t)W_1(t) + P_2(t)W_2(t)] + k_2\Theta(t) - k_3C(t) \tag{28}$$

Due to fixed academic program duration, a **Non-linear Constraint** is placed on the total achievable competence. A permissive Gen-AI policy (ϵ) acts as a **productivity shifter**, allowing for higher skill density:

5.2 Tailored Policy Recommendations for Three Archetypes

Based on the optimization results, we propose a triad framework encompassing *Scale*, *Mode*, and *Content*, customized for three distinct professional archetypes.

It is worth noting that the cultivation expectations for comprehensive qualities weakly related to careers are 82.5, 82.49, and 82.4, respectively. Therefore, schools need to pay attention to cultivating students' comprehensive abilities such as aesthetic education, philosophy, and social skills in their curriculum design.

5.2.1 Actuarial Science

We select the Bachelor of Science in Actuarial Science program at The Ohio State University for analysis.

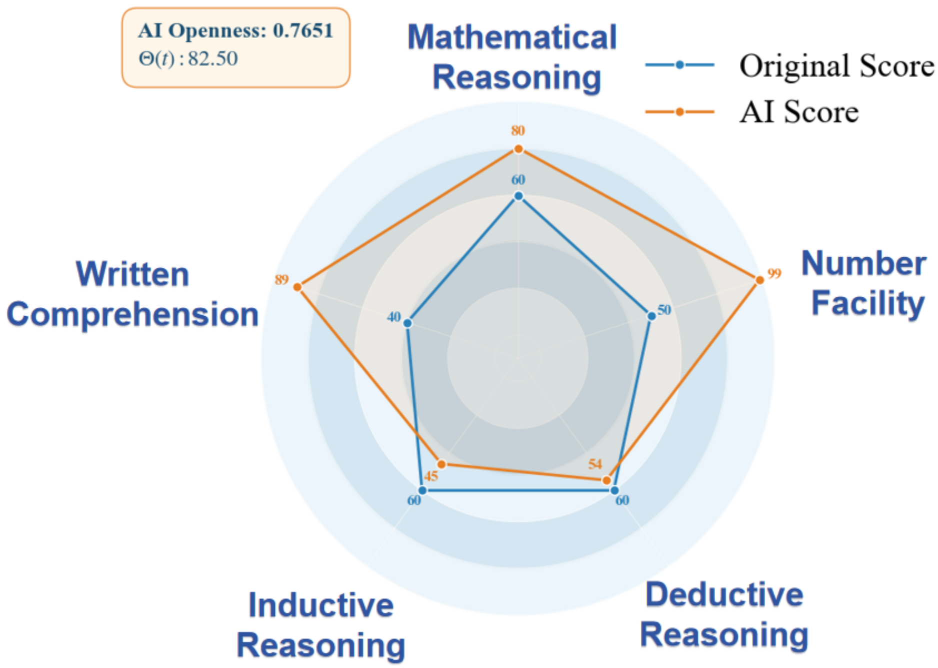


Figure 10: Radar Map of Actuarial Science

The chart illustrates a profound Gen-AI-driven capability expansion in quantitative and communicative throughput under high Gen-AI openness (0.7651). However, the plateau in core inductive and deductive reasoning highlights a critical cognitive offloading risk, where foundational logic is bypassed for machine-enabled efficiency. Therefore, we have formulated the following targeted educational reform policies.

A. Strategic Scale: Elite Expansion Given the rising demand for Gen-AI-empowered actuaries, we recommend Strategic Expansion with Elite Selection.

B. Pedagogical Mode: High-Openness Sandbox: We propose a High-Openness Gen-AI model ($\epsilon = 0.7651$),allowing students to use Gen-AI assistance when dealing with complex mathematical proofs and large-scale data modeling.

C. Curricular Architecture: Skill-Biased Restructuring

1. **De-emphasize Pure Calculation:** The Gen-AI enhancement for Number Facility ranges from 50 to 99. The Math 3618 course design should focus on assessing students' Deductive Reasoning.

2. **Restructure Programming Courses:** CSE 2111 should shift from basic Excel formulas to Gen-AI-driven automated data cleaning and predictive modeling.

5.2.2 Culinary Arts

We select the Associate of Occupational Studies (AOS) Degree in Culinary Arts program at the Auguste Escoffier School of Culinary Arts for analysis.

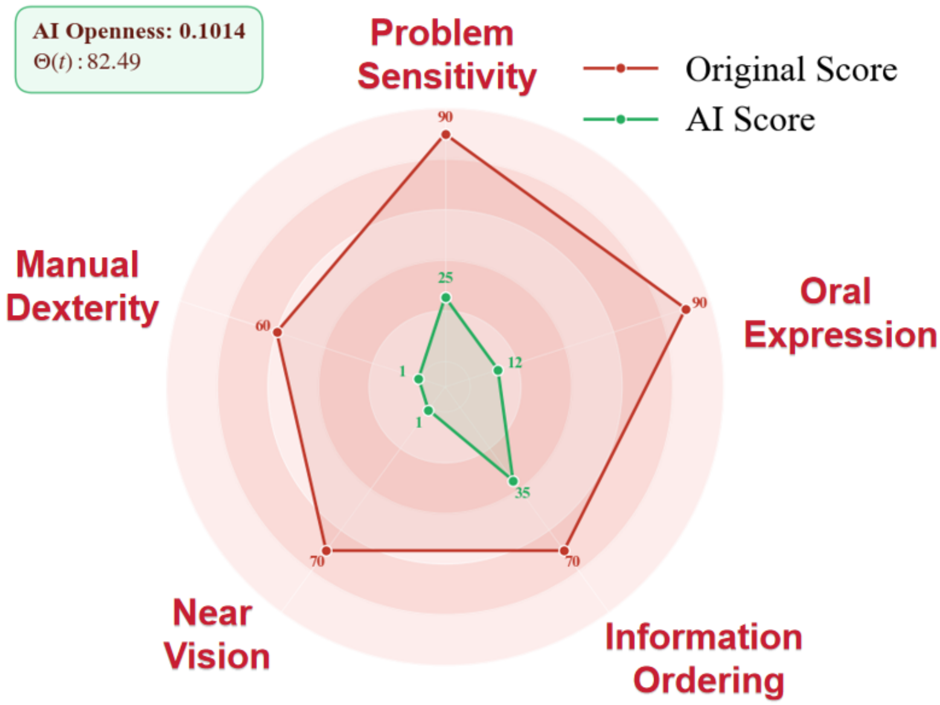


Figure 11: Radar Map of Culinary Arts

The chart reveals negligible Gen-AI synergy, as core physical dexterity remains immune to machine enhancement. This justifies the highly restricted openness (0.1014) to prevent cognitive off-loading and preserve traditional craftsmanship. Therefore, we have formulated the following targeted policies.

A. Scale Strategy: Maintain With slow demand growth for Gen-AI-integrated chefs and the continued dominance of traditional roles, we recommend stable or slightly contracted enrollment to maintain small-class instruction.

B. Pedagogical Mode: Restricted Gen-AI Governance. We advocate for a Strictly Restricted Gen-AI policy ($\epsilon = 0.1014$). Over-reliance on Gen-AI diminishes a chef's capacity for real-time decision-making and coordination in complex kitchen environments.

C. Curricular Content: Mastery of Tradition and Narrative

1. **Sensory and Physical Core:** Gen-AI intervention should be prohibited in courses

like Culinary Foundations to strengthen knife skills and sensory training that Gen-AI cannot replace.

2. **Operations-Focused Gen-AI:** Gen-AI utility should be confined to back-end subjects like Culinary Math and Accounting for supply chain and food safety management.

5.2.3 Graphic Design

We select the Bachelor of Fine Arts (BFA) in Graphic Design program at Kendall College of Art and Design of Ferris State University for analysis.

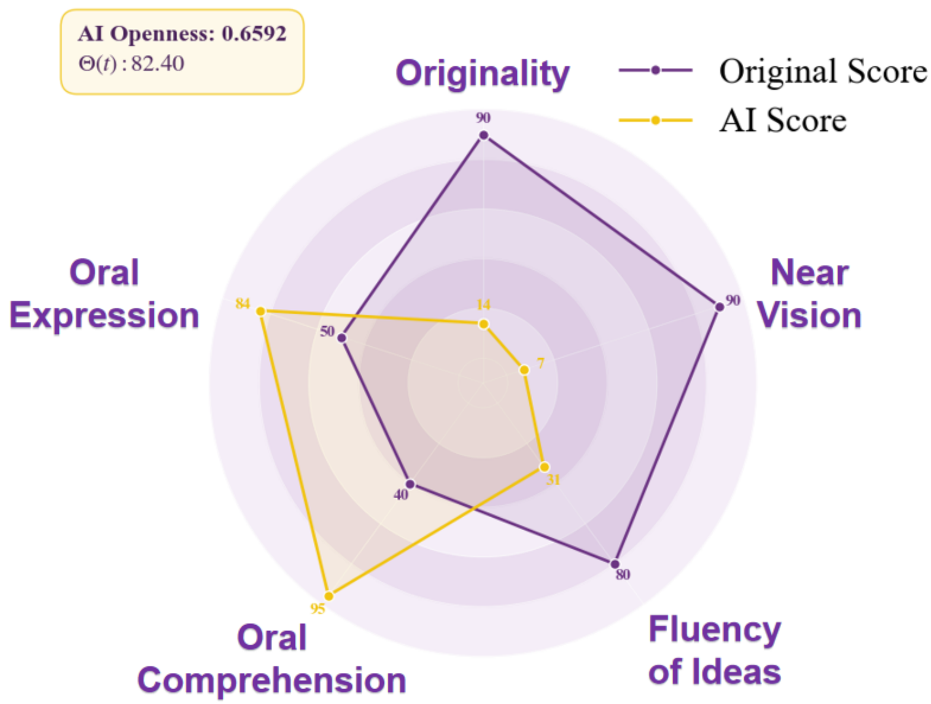


Figure 12: Radar Map of Graphic Design

The chart highlights a sharp gap: Gen-AI excels in execution but remains deficient in Originality. This necessitates a strategic shift toward high-level curation at 0.6592 openness to leverage machine efficiency without losing creative intent. Therefore, we have formulated the following targeted policies.

A. Scale Strategy: Structural Optimization As demand for Gen-AI-integrated roles fluctuates, we recommend downsizing the program to reallocate resources toward protecting original human creativity and high-end talent transformation.

B. Pedagogical Mode: Creative Collaboration Model. With an Gen-AI openness of 0.6592, we propose a non-linear intervention strategy. Low-level courses should remain low-AI to build aesthetic intuition, while senior projects should integrate Gen-AI for verbal comprehension and expression.

C. Curricular Content: Curation and Ethical Synthesis

1. **Strengthen "Manual Quality Control":** In Typography courses, the focus should be on assessing details (Near Vision) and original logic that Gen-AI cannot match.
2. **Introduce "AI Communication Officer":** Utilize Gen-AI's strong Oral Comprehension to assist with client proposals and business copywriting in Professional Practices.

6 Sensitivity Analysis

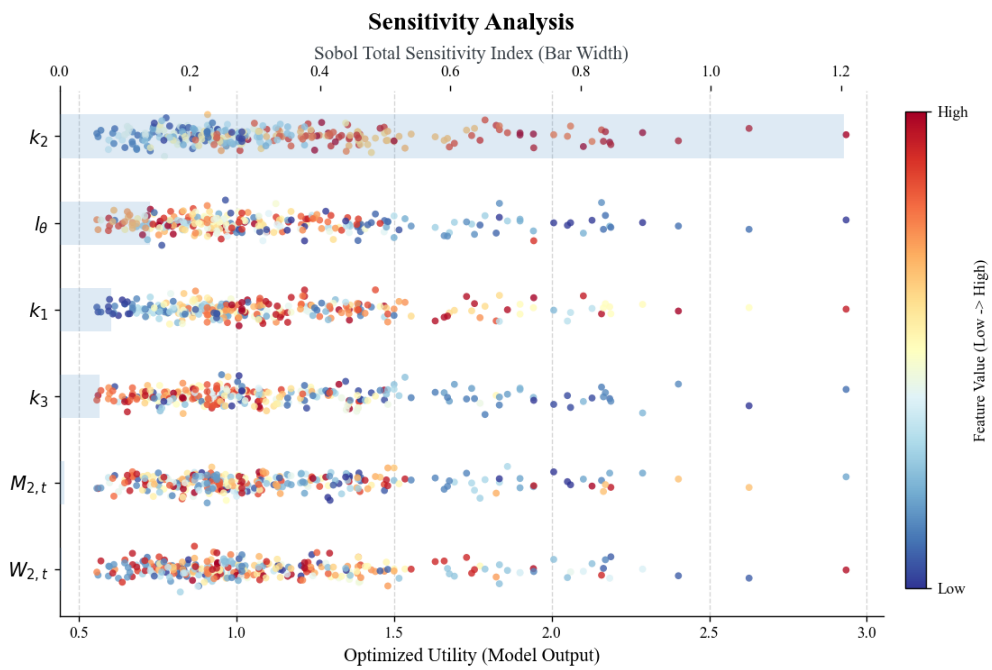


Figure 13: Sensitivity Analysis

The **Sobol global sensitivity analysis** was conducted using quasi-random sampling to quantify parameter influence via the Total Sensitivity Index (S_T). Results for the six most sensitive parameters show that all S_T values are notably low ($\max \approx 0.26$), indicating that output fluctuations are not drastically amplified by single-parameter perturbations. This confirms the **high robustness and numerical stability** of the optimization model under complex parameter conditions.

7 Further Discussion on Our Models

7.1 External Constraints on Gen-AI Advancement

The effective Gen-AI capability $E\vec{a}(t)$ is limited by real-world constraints:

- **Carbon Footprint / Compute Limits:** Compute-heavy Gen-AI domains (e.g., advanced **Reasoning** for actuaries) face a hard limit governed by **Token-Efficiency** and **Carbon Footprint**.

  $\Rightarrow$ **Adjustment:** Introduce a *damping factor* λ_{CF} on the $\vec{a}(t)$ vector for computationally intensive tasks.

- **Intellectual Property (IP) Crisis:** For visual professions like Graphic Design, the crisis of **Attribution** and **Intellectual Property** increases the risk of commercial Gen-AI adoption.

  $\Rightarrow$ **Adjustment:** Introduce a *downward adjustment* λ_{IP} to the effective Augmentation Index $\vec{z}(t)$ for commercially-used Gen-AI outputs.

7.2 Generalizability of the CFE and EPF Models

The core logic of our model, like **Substitution-Augmentation dynamics** and **Skill-Biased Wage Premiums**, can be applied to a wide range of knowledge and service-intensive industries by simply calibrating the task weights ($\vec{w}$) and sectoral sensitivities (ψ, δ). So it exhibits strong generalizability tested across the full spectrum of Gen-AI-impact archetypes.

The model is primarily applicable within two boundaries:

- **Non-AI Frontier Occupations:** Excludes sectors whose primary function is the direct development of Gen-AI technology itself (e.g., Gen-AI researchers).
- **Stable Traditional Demand:** Requires that the occupation's overall market growth (ρ) is stable, ensuring the model focuses on **internal skill reallocation** rather than total market collapse.

7.3 Comparative Analysis of Future Gen-AI Scenarios

As mentioned in the JDAT Model, the trajectory of Gen-AI's evolution is dictated by the interplay between the acceleration factor α and decay factor δ:

- **Exponential Expansion ($\alpha > \delta$):** Gen-AI capabilities accelerate indefinitely. PSIs must prioritize **"Agile Adaptation"** (e.g., micro-credentials) to keep pace.
- **Asymptotic Convergence ($\delta \geq \alpha$):** Diminishing returns (**Data Wall**) dominate. PSIs should emphasize **"Deep Engineering Proficiency"** (optimizing mature models) over chasing novelty.

7.4 Emergent Gen-AI Niches: New Job Creation

Inspired by the **Turing Trap**, we notice that Gen-AI may not only restructure current jobs but also create entirely new sectors with high **Industrial Rebirth Coefficients (IRC)**. Here are two examples of such new occupational niches:

- **Synthetic Data Manufacturing:** Creates a data-supply loop to counter **Model Degradation** ("Data Rot").
- **Algorithmic Liability & Gen-AI Auditing:** Insures against Gen-AI output risks (e.g., **Hallucination Risk Cost**).

Consequently, we recommend PSIs to look beyond simple vocational upskilling and establish specialized **"Emergent Gen-AI Niches" Curricula** to strategically occupying potential, high-value labor market segments.

8 Model Evaluation

1.Strength

- **High Fidelity & Systemic Logic:** Integrates the **Task-Based Approach** with **System Dynamics**. This captures non-linear interdependencies and feedback loops.
- **Uncertainty Quantification:** Forecasts include **Confidence Intervals**. **Scenario Analysis** enables flexible, "no-regret" policy design against stochastic risk.
- **Multi-Scale Integration:** Bridges micro-level task evolution (CFE Model) with macro-level labor shifts. The EPF Model incorporates **Cost-Benefit Analysis**.
- **Robustness:** The framework is adaptable across diverse sectors. **Sensitivity Analysis** confirms output stability.

2.Weakness

- **Data Dependency:** Predictive accuracy is critically contingent on the authenticity, granularity, and timeliness of input datasets.
- **Local Abstraction:** The model simplifies **geographical segmentation** and **cultural divergence** in policy discussions for specific institutions.
- **Excessive Linearization:** The model relies heavily on linear assumptions. While these are reliable within small fluctuation ranges, they inevitably present a discrepancy with complex real-world dynamics.

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